

MIT Integration Bee Qualifying Exam 2023 Problem 5

Problem:

$$\int_{-\pi/2}^{\pi/2} \cos(20x) \sin(25x) dx$$

Solution:

Solution. Let

$$f(x) = \cos(20x) \sin(25x).$$

We check the parity of the integrand:

$$f(-x) = \cos(20(-x)) \sin(25(-x)).$$

Using

$$\cos(-u) = \cos(u), \quad \sin(-u) = -\sin(u),$$

we get

$$f(-x) = \cos(20x)(-\sin(25x)).$$

Thus

$$f(-x) = -\cos(20x) \sin(25x) = -f(x).$$

Therefore, $f(x)$ is an odd function. Since the interval

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

is symmetric about 0, the integral of an odd function over this interval is 0. Hence,

$$\int_{-\pi/2}^{\pi/2} \cos(20x) \sin(25x) dx = \boxed{0}.$$

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